

AI-specific observability tools add another layer of visibility. Langfuse enables prompt tracing and response analysis, while Arize Phoenix helps evaluate retrieval effectiveness and response quality. Together, these tools provide end-to-end visibility into both system performance and AI behaviour.

By combining infrastructure monitoring, distributed tracing, and AI-specific evaluation, organisations can quickly identify bottlenecks, diagnose failures, optimise costs, and improve response quality across their AI workloads.

Example use case: Monitoring an enterprise RAG assistant

An enterprise knowledge assistant can help workers locate internal documents, policies, and technical guides. The assistant in our example is based on an RAG architecture, which enriches user queries with relevant information from a vector database before sending them to a large language model.

When a user submits a question, the request first goes through the orchestration layer. The system retrieves the most relevant documents from the knowledge base and constructs a prompt that consists of the user query and the context extracted from the retrieved documents. The language model generates a response and returns it to the user.

Meanwhile, observability tools are constantly collecting telemetry data. OpenTelemetry traces the request through the application components, and Prometheus collects metrics such as request volume, latency, and resource utilisation. Grafana dashboards offer real-time insight into system performance.

AI-specific observability platforms provide more profound insights. Langfuse logs prompts,

retrieved context, model responses, and token usage. Arize Phoenix helps in evaluating the quality of retrieval and finding cases where the retrieved documents were not sufficient or not relevant. OpenLIT provides visibility into the latency of the model, the cost of the APIs, and the performance of the entire AI workload.

If response latency is increasing, retrieval quality is decreasing, or token consumption is unexpectedly spiking, engineering teams can quickly identify the root cause and take corrective action. This visibility helps keep the assistant reliable, cost-effective, and useful for end users.

Benefits of AI observability

There are many practical benefits of implementing observability across AI workloads and LLM applications for engineering teams and organisations.

- **Enhanced reliability:** Teams can swiftly identify failures within retrieval pipelines, model interactions, and support services before they impact users.
- **Faster troubleshooting:** End-to-end traces help engineers identify bottlenecks and troubleshoot issues faster.
- **Improved response quality:** Monitoring retrieval effectiveness, hallucinations, and user feedback improves the accuracy and relevance of generated responses.
- **Cost optimization:** Companies

gain visibility into token usage, inference expenses, and resource utilisation, helping them manage operating costs.

- **Enhanced user experience:** Tracking latency, response quality, and satisfaction metrics ensures consistent and reliable AI interactions.
- **Improved operational visibility:** A single observability framework provides teams with a comprehensive view of infrastructure, applications, AI workloads, and model behaviour.

AI and LLM applications are rapidly becoming a foundational element in modern enterprise technology stacks. But operating these systems demands more than traditional infrastructure monitoring. AI workloads pose extra challenges and are often difficult to understand and troubleshoot with traditional methods of observability.

AI observability is therefore a critical discipline for organisations deploying production-grade AI solutions. By using infrastructure monitoring, distributed tracing, AI-specific data collection, and quality checks, teams can see everything happening in the entire AI process. Organisations that invest in observability today will be better placed to scale their AI initiatives successfully in the years to come. **END** 

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